Subnetting Practical: 192.168.10.0/24

Question 1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

Step 1: Find the new prefix. Four subnets are needed. Borrow 2 host bits because $2^2 = 4$. Add 2 to the original /24 prefix: /24 + 2 = **/26**.

Step 2: Convert /26 to a subnet mask. The first 24 bits are 255.255.255.0. The last octet has two network bits: 11000000 in binary = 128 + 64 = 192. So the mask is **255.255.255.192**.

Step 3: Find the block size. Block size = 256 – 192 = **64**. Network addresses therefore increase by 64 in the last octet: 0, 64, 128, 192.

Subnet	Network address	Subnet mask	Broadcast address	Usable host range
1	192.168.10.0/26	255.255.255.192	192.168.10.63	192.168.10.1–192.168.10.62
2	192.168.10.64/26	255.255.255.192	192.168.10.127	192.168.10.65–192.168.10.126
3	192.168.10.128/26	255.255.255.192	192.168.10.191	192.168.10.129–192.168.10.190
4	192.168.10.192/26	255.255.255.192	192.168.10.255	192.168.10.193–192.168.10.254

Answer: Each /26 subnet has 64 total addresses: 1 network address, 62 usable host addresses, and 1 broadcast address.

Question 2. For one subnet, assign valid IP addresses to 5 devices and list which IP goes to which device.

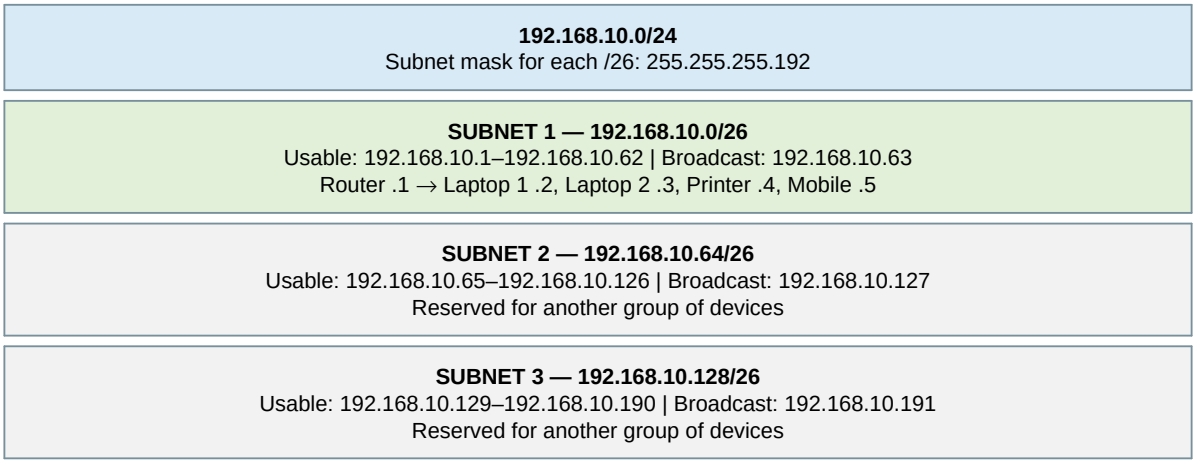
Use Subnet 1: **192.168.10.0/26**. Its network address is .0, broadcast address is .63, and usable host range is .1–.62. The five assigned addresses below are unique and valid.

Device	Assigned IP	Subnet mask	Default gateway
Router (LAN)	192.168.10.1	255.255.255.192	N/A — it is the gateway
Laptop 1	192.168.10.2	255.255.255.192	192.168.10.1
Laptop 2	192.168.10.3	255.255.255.192	192.168.10.1
Printer	192.168.10.4	255.255.255.192	192.168.10.1
Mobile	192.168.10.5	255.255.255.192	192.168.10.1

Important: Do not assign .0 (network address) or .63 (broadcast address) to a device. In a real home network, check the router's DHCP pool before using static addresses so there are no conflicts.

Question 3. Draw a simple diagram showing the subnet plan, IP ranges, and where each device is connected.

This logical diagram shows the four /26 subnets. The five sample devices are on Subnet 1. Subnets 2–4 are reserved for other device groups.



SUBNET 4 — 192.168.10.192/26
Usable: 192.168.10.193–192.168.10.254 | Broadcast: 192.168.10.255
Reserved for another group of devices

Connection note: Router LAN (192.168.10.1) connects to a switch or router LAN ports; Laptop 1, Laptop 2, and Printer can connect by Ethernet, while Mobile usually connects by Wi-Fi. This is a logical addressing diagram. To keep all four subnets separate in a real network, configure VLANs and routing or separate router interfaces; assigning different IP ranges alone does not automatically separate a flat Wi-Fi LAN.

Question 4. Explain step by step how you calculated the subnet mask, number of hosts, and assigned IP addresses.

Step 1 — Start with the original network. 192.168.10.0/24 means the first 24 bits are the network portion and 8 bits are available for hosts. Its original mask is 255.255.255.0.

Step 2 — Borrow bits for four subnets. The formula is 2■, where n is the number of borrowed bits. Borrowing 2 bits gives $2^2 = 4$ subnets, exactly what is required.

Step 3 — Calculate the new prefix. Add the borrowed bits to /24: 24 + 2 = /26. Convert /26 to dotted decimal: 255.255.255.192.

Step 4 — Calculate host addresses. There are 32 − 26 = 6 host bits left. Total addresses per subnet = 2■ = 64. Subtract the network and broadcast addresses: 64 − 2 = **62 usable host addresses** per subnet.

Step 5 — Find the block size. Take 256 minus the mask's last octet: 256 − 192 = 64. Count by 64 to get network IDs: .0, .64, .128, and .192.

Step 6 — Calculate each broadcast address. The broadcast is the last address before the next subnet begins. The first subnet ends at .63, the second at .127, the third at .191, and the fourth at .255.

Step 7 — Find usable ranges. The first host is network + 1; the last host is broadcast − 1. For example, Subnet 1 uses .1 through .62.

Step 8 — Assign device addresses. Assign router .1, Laptop 1 .2, Laptop 2 .3, Printer .4, and Mobile .5. These are unique, lie in the same /26 subnet, and are not the network or broadcast address. Use mask 255.255.255.192 on each device and gateway 192.168.10.1 on end devices.

Step 9 — Verify. Check that each device has a different IP and the same correct mask. After configuring the network, test with ping. The router/VLAN configuration must support the subnet plan if multiple subnets are actually being used.